

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250286460

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Nogawa; Masashi

MULTI-PHASE VOLTAGE REGULATOR OPERABLE WITH FEWER FEEDBACK LOOPS

Abstract

A multi-phase voltage regulator operable with fewer feedback loops is described in the present disclosure. The multi-phase voltage regulator is configured to generate an output current to charge an output capacitor to thereby provide an output voltage. As the name suggests, the output current includes multiple sub-currents each generated by a respective one of multiple sub-current generators. Herein, each of the sub-current generators is configured to self-regulate a respective one of the sub-currents based on an expected level of the output voltage such that each of the sub-currents is substantially identical to each of the other sub-currents. By letting each sub-current generator self-regulate its own sub-current, it is possible to reduce the number of feedback loops in the multi-phase voltage regulator. As a result, the multi-phase voltage regulator can operate with a faster transient response, wider loop bandwidth, increased phase/gain margin, and higher control loop stability.

Inventors: Nogawa; Masashi (Sachse, TX)

Applicant: Qorvo US, Inc. (Greensboro, NC)

Family ID: 1000008463622

Appl. No.: 19/046624

Filed: February 06, 2025

Related U.S. Application Data

us-provisional-application US 63562760 20240308

us-provisional-application US 63656285 20240605

Publication Classification

Int. Cl.: H02M3/158 (20060101); H02M1/00 (20070101)

Background/Summary

RELATED APPLICATIONS [0001] This application claims the benefit of U.S. provisional patent application Ser. No. 63/562,760, filed on Mar. 8, 2024, and U.S. provisional patent application Ser. No. 63/656,285, filed on Jun. 5, 2024, the disclosures of which are hereby incorporated herein by reference in their entireties.

FIELD OF THE DISCLOSURE

[0002] The technology of the disclosure relates generally to a multi-phase DC-DC voltage regulator in an electronic power system.

BACKGROUND

[0003] Electronic power systems are the enabling infrastructure technology that promotes conversion and distribution of electrical power from a power source to electronics and electrical machines. A power conversion circuit is often at the heart of each electronic power system converting electrical power from raw form and quantity as produced by the power source to an appropriate form and quantity as needed by machines, motors, electronic equipment, and so on.

[0004] DC-DC conversion has always been an integral element of switch-mode power supplies that operate by toggling a main switch between on—(a.k.a. closed) and off—(a.k.a. open) states. More specifically, the DC-DC conversion can be carried out by a buck (a.k.a. step-down) converter or a boost (a.k.a. step-up) converter. The buck converter passes energy directly to an output with an energy storage inductor providing a continuing current to the output when the main switch is in the off-state, whereas the boost converter stores all the output energy in an inductor when the main switch is in the on-state and passes the stored energy to the output when the main switch is in the off-state.

SUMMARY

[0005] Aspects disclosed in the detailed description are related to a multi-phase voltage regulator operable with fewer feedback loops. The multi-phase voltage regulator is configured to generate an output current to charge an output capacitor to thereby provide an output voltage. As the name suggests, the output current includes multiple sub-currents each generated by a respective one of multiple sub-current generators. Herein, each of the sub-current generators is configured to self-regulate a respective one of the sub-currents based on an expected level of the output voltage such that each of the sub-currents is substantially identical to each of the other sub-currents. By letting each sub-current generator self-regulate its own sub-current, it is possible to reduce the number of feedback loops in the multi-phase voltage regulator. As a result, the multi-phase voltage regulator can operate with a faster transient response, wider loop bandwidth, increased phase/gain margin, and higher control loop stability.

[0006] In one aspect, a multi-phase voltage regulator is provided. The multi-phase voltage regulator includes an output capacitor. The output capacitor is coupled to a voltage output and charged by an output current, which includes multiple sub-currents, to provide an output voltage at the voltage output. The multi-phase voltage regulator also includes a voltage control circuit. The voltage control circuit is configured to receive a feedback voltage indicating a present level of the output voltage at the voltage output. The voltage control circuit is also configured to determine a target voltage indicating an expected level of the output voltage based on the feedback voltage and a reference voltage. The multi-phase voltage regulator also includes multiple sub-current generators. Each of the multiple sub-current generators is coupled to the voltage output. Each of the

multiple sub-current generators is configured to receive the target voltage from the voltage control circuit. Each of the multiple sub-current generators is configured to extrapolate the expected level of the output current at the voltage output from the target voltage. Each of the multiple sub-current generators is configured to regulate the respective one of the multiple sub-currents based on the expected level of the output current such that the respective one of the multiple sub-currents is substantially identical to each of the other multiple sub-currents comprised in the output current.

[0007] In another aspect, a method for operating a multi-phase voltage regulator with fewer feedback loops is provided. The method includes charging an output capacitor by an output current comprising multiple sub-currents to provide an output voltage at a voltage output. The method also includes receiving a feedback voltage indicating a present level of the output voltage at the voltage output. The method also includes determining a target voltage indicating an expected level of the output voltage based on the feedback voltage and a reference voltage. The method also includes extrapolating the expected level of the output current at the voltage output from the target voltage. The method also includes regulating each of the multiple sub-currents based on the expected level of the output current such that the respective one of the multiple sub-currents is substantially identical to each of the other multiple sub-currents comprised in the output current.

[0008] In another aspect, an electronic power system is provided. The electronic power system includes a conversion circuit. The conversion circuit is coupled between a power source and a load circuit. The conversion circuit includes a multi-phase voltage regulator. The multi-phase voltage regulator includes an output capacitor. The output capacitor is coupled to a voltage output and charged by an output current, which includes multiple sub-currents, to provide an output voltage at the voltage output. The multi-phase voltage regulator also includes a voltage control circuit. The voltage control circuit is configured to receive a feedback voltage indicating a present level of the output voltage at the voltage output. The voltage control circuit is also configured to determine a target voltage indicating an expected level of the output voltage based on the feedback voltage and a reference voltage. The multi-phase voltage regulator also includes multiple sub-current generators. Each of the multiple sub-current generators is coupled to the voltage output. Each of the multiple sub-current generators is configured to receive the target voltage from the voltage control circuit. Each of the multiple sub-current generators is configured to extrapolate the expected level of the output current at the voltage output from the target voltage. Each of the multiple sub-current generators is configured to regulate the respective one of the multiple sub-currents based on the expected level of the output current such that the respective one of the multiple sub-currents is substantially identical to each of the other multiple sub-currents comprised in the output current.

[0009] Those skilled in the art will appreciate the scope of the disclosure and realize additional aspects thereof after reading the following detailed description in association with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The accompanying drawings incorporated in and forming a part of this specification illustrate several aspects of the disclosure and, together with the description, serve to explain the principles of the disclosure.

[0011] FIG. 1 is a schematic diagram of an exemplary existing multi-phase voltage regulator operable with three feedback loops;

[0012] FIG. 2 is a schematic diagram of an exemplary multi-phase voltage regulator configured according to an embodiment of the present disclosure to operate with fewer feedback loops than the existing multi-phase voltage regulator of FIG. 1;

[0013] FIG. 3 is a schematic diagram of an exemplary voltage control circuit in the multi-phase

voltage regulator of FIG. 2;

[0014] FIG. 4 is a graphic diagram illustrating a load line curve employed by the voltage control circuit of FIG. 3A;

[0015] FIG. 5 is a schematic diagram of an exemplary current determination circuit that can be provided in the multi-phase voltage regulator of FIG. 2 to determine and report an output current in the multi-phase voltage regulator of FIG. 2;

[0016] FIG. 6 is a schematic diagram of an exemplary electronic power system wherein the multi-phase voltage regulator of FIG. 2 can be provided; and

[0017] FIG. 7 is a flowchart of an exemplary process whereby the multi-phase voltage regulator of FIG. 2 can be configured to operate with fewer feedback loops.

DETAILED DESCRIPTION

[0018] The embodiments set forth below represent the necessary information to enable those skilled in the art to practice the embodiments and illustrate the best mode of practicing the embodiments. Upon reading the following description in light of the accompanying drawing figures, those skilled in the art will understand the concepts of the disclosure and will recognize applications of these concepts not particularly addressed herein. It should be understood that these concepts and applications fall within the scope of the disclosure and the accompanying claims.

[0019] It will be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from the scope of the present disclosure. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

[0020] It will be understood that when an element such as a layer, region, or substrate is referred to as being “on” or extending “onto” another element, it can be directly on or extend directly onto the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly on” or extending “directly onto” another element, there are no intervening elements present. Likewise, it will be understood that when an element such as a layer, region, or substrate is referred to as being “over” or extending “over” another element, it can be directly over or extend directly over the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly over” or extending “directly over” another element, there are no intervening elements present. It will also be understood that when an element is referred to as being “connected” or “coupled” to another element, it can be directly connected or coupled to the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected” or “directly coupled” to another element, there are no intervening elements present.

[0021] Relative terms such as “below” or “above” or “upper” or “lower” or “horizontal” or “vertical” may be used herein to describe a relationship of one element, layer, or region to another element, layer, or region as illustrated in the Figures. It will be understood that these terms and those discussed above are intended to encompass different orientations of the device in addition to the orientation depicted in the Figures.

[0022] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the disclosure. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises,” “comprising,” “includes,” and/or “including” when used herein specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0023] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this

disclosure belongs. It will be further understood that terms used herein should be interpreted as having a meaning that is consistent with their meaning in the context of this specification and the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

[0024] Aspects disclosed in the detailed description are related to a multi-phase voltage regulator operable with fewer feedback loops. The multi-phase voltage regulator is configured to generate an output current to charge an output capacitor to thereby provide an output voltage. As the name suggests, the output current includes multiple sub-currents each generated by a respective one of multiple sub-current generators. Herein, each of the sub-current generators is configured to self-regulate a respective one of the sub-currents based on an expected level of the output voltage such that each of the sub-currents is substantially identical to each of the other sub-currents. By letting each sub-current generator self-regulate its own sub-current, it is possible to reduce the number of feedback loops in the multi-phase voltage regulator. As a result, the multi-phase voltage regulator can operate with a faster transient response, wider loop bandwidth, increased phase/gain margin, and higher control loop stability.

[0025] Before discussing a multi-phase voltage regulator of the present disclosure, starting at FIG. 2, a brief overview of an existing multi-phase voltage regulator is first provided with reference to FIG. 1 to help identify the technical problems to be solved herein.

[0026] FIG. 1 is a schematic diagram of an exemplary existing multi-phase voltage regulator **10** operable based on a set of first feedback loops Loop**1**(**1**)-Loop**1**(**N**), a second feedback loop Loop**2**, and a set of third feedback loops Loop**3**(**1**)-Loop**3**(**N**). The existing multi-phase voltage regulator **10** includes an output capacitor C.sub.OUT. Herein, the output capacitor C.sub.OUT is coupled to a voltage output **12** and charged by an output current I.sub.OUT to provide an output voltage V.sub.OUT at the voltage output **12**.

[0027] As the name “multi-phase voltage regulator” suggests, the output current I.sub.OUT is a multi-phase current that includes multiple sub-currents I.sub.OUT-1-I.sub.OUT-N ($I_{\text{sub.OUT}} = \sum_{i=1}^N I_{\text{sub.OUT-}i}$). Each of the sub-currents I.sub.OUT-1-I.sub.OUT-N is generated by a respective one of multiple sub-current generators **14**(**1**)-**14**(**N**). Specifically, each of the sub-current generators **14**(**1**)-**14**(**N**) includes a respective one of multiple current regulators **16**(**1**)-**16**(**N**) and a respective one of multiple current sensing circuits **18**(**1**)-**18**(**N**).

[0028] Each of the current sensing circuits **18**(**1**)-**18**(**N**) is part of a respective one of the set of first feedback loops Loop**1**(**1**)-Loop**1**(**N**) and configured to generate a respective one of multiple sense currents I.sub.SNS-1-I.sub.SNS-N. Each of the sense currents I.sub.SNS-1-I.sub.SNS-N is generated to indicate a present level of a respective one of the sub-currents I.sub.OUT-1-I.sub.OUT-N at the voltage output **12**. In an embodiment, each of the sense currents I.sub.SNS-1-I.sub.SNS-N may be proportionally related to the respective one of the sub-currents I.sub.OUT-1-I.sub.OUT-N based on a scaling factor, which can be less than or equal to one.

[0029] The existing multi-phase voltage regulator **10** also includes a voltage control circuit **20**. The voltage control circuit **20** is configured to receive a feedback voltage V.sub.OUT-FB via the second feedback loop Loop**2**. The feedback voltage V.sub.OUT-FB provides an indication of the present level of the output voltage V.sub.OUT at the voltage output **12**. The voltage control circuit **20** is configured to compare the feedback voltage V.sub.OUT-FB against a reference voltage V.sub.REF to thereby determine a target voltage V.sub.TGT that indicates an expected level of the output voltage V.sub.OUT. The voltage control circuit **20** then provides the target voltage V.sub.TGT to each of the current regulators **16**(**1**)-**16**(**N**). Each of the current regulators **16**(**1**)-**16**(**N**), in turn, generates the respective one of the sub-currents I.sub.OUT-1-I.sub.OUT-N in accordance with the target voltage V.sub.TGT and the respective one of the sense currents I.sub.SNS-1-I.sub.SNS-N.

[0030] In the existing multi-phase voltage regulator **10**, it is desirable for the sub-currents I.sub.OUT-1-I.sub.OUT-N to be substantially identical to each other. To balance the sub-currents I.sub.OUT-1-I.sub.OUT-N across the sub-current generators **14**(**1**)-**14**(**N**), a current balancing

circuit **22** is provided in the existing multi-phase voltage regulator **10**. The current balancing circuit **22** is configured to receive the sense currents $I_{\text{sub.SNS-1}}-I_{\text{sub.SNS-N}}$ via the set of third feedback loops $\text{Loop3(1)}-\text{Loop3(N)}$, respectively, and generate a target current $I_{\text{sub.OUT-TGT}}$ to indicate an expected sum of the sub-currents $I_{\text{sub.OUT-1}}-I_{\text{sub.OUT-N}}$. Accordingly, the current balancing circuit **22** can cause each of the sub-currents $I_{\text{sub.OUT-1}}-I_{\text{sub.OUT-N}}$ to be adjusted to an equal amount.

[0031] Notably, the current balancing circuit **22** and the set of third feedback loops $\text{Loop3(1)}-\text{Loop3(N)}$ can lead to a more complex implementation and potentially hinder the ability of the existing multi-phase voltage regulator **10** to respond quickly to transient events. As such, it is desirable to eliminate the current balancing circuit **22** and the set of third feedback loops $\text{Loop3(1)}-\text{Loop3(N)}$ to help improve the response time of the existing multi-phase voltage regulator **10**.

[0032] In this regard, FIG. **2** is a schematic diagram of an exemplary multi-phase voltage regulator **24** configured according to an embodiment of the present disclosure to operate with only a set of first feedback loops $\text{Loop1(1)}-\text{Loop1(N)}$ and a second feedback loop **Loop2**. As described below, by eliminating the set of third feedback loops $\text{Loop3(1)}-\text{Loop3(N)}$ and reconfiguring the voltage control circuit **20** in the existing multi-phase voltage regulator **10** of FIG. **1**, the multi-phase voltage regulator **24** is able to provide a faster response to the transient event than the existing multi-phase voltage regulator **10**.

[0033] Like the existing multi-phase voltage regulator **10**, the multi-phase voltage regulator **24** also includes an output capacitor $C_{\text{sub.OUT}}$. The output capacitor $C_{\text{sub.OUT}}$ is coupled to a voltage output **26** and charged by an output current $I_{\text{sub.OUT}}$ to provide an output voltage $V_{\text{sub.OUT}}$ at the voltage output **26**.

[0034] Like the output current $I_{\text{sub.OUT}}$ in the existing multi-phase voltage regulator **10**, the output current $I_{\text{sub.OUT}}$ is also a multi-phase current that includes multiple sub-currents $I_{\text{sub.OUT-1}}-I_{\text{sub.OUT-N}}$ ($I_{\text{sub.OUT}}=\sum_{i=1}^{N} I_{\text{sub.OUT-i}}$). Each of the sub-currents $I_{\text{sub.OUT-1}}-I_{\text{sub.OUT-N}}$ is generated by a respective one of multiple sub-current generators **28(1)-28(N)**. Specifically, each of the sub-current generators **28(1)-28(N)** includes a respective one of multiple current regulators **30(1)-30(N)** and a respective one of multiple current sensing circuits **32(1)-32(N)**.

[0035] Each of the current sensing circuits **32(1)-32(N)** is associated with a current sense gain $G_{\text{sub.IS}}$ and is part of a respective one of the set of first feedback loops $\text{Loop1(1)}-\text{Loop1(N)}$. Similar to the current sensing circuits **18(1)-18(N)**, each of the current sensing circuits **32(1)-32(N)** is also configured to generate a respective one of multiple sense currents $I_{\text{sub.SNS-1}}-I_{\text{sub.SNS-N}}$ that indicates a present level of a respective one of the sub-currents $I_{\text{sub.OUT-1}}-I_{\text{sub.OUT-N}}$ at the voltage output **26**. Herein, each of the sense currents $I_{\text{sub.SNS-1}}-I_{\text{sub.SNS-N}}$ may be proportionally related to the respective one of the sub-currents $I_{\text{sub.OUT-1}}-I_{\text{sub.OUT-N}}$ based on a scaling factor, which can be less than or equal to one.

[0036] The multi-phase voltage regulator **24** also includes a voltage control circuit **34**. The voltage control circuit **34** is configured to receive a feedback voltage $V_{\text{sub.OUT-FB}}$ via the second feedback loop **Loop2** (a.k.a. output voltage feedback loop). The feedback voltage $V_{\text{sub.OUT-FB}}$ provides an indication of the present level of the output voltage $V_{\text{sub.OUT}}$ at the voltage output **26**. The voltage control circuit **34** is configured to compare the feedback voltage $V_{\text{sub.OUT-FB}}$ against a reference voltage $V_{\text{sub.REF}}$ to thereby determine a target voltage $V_{\text{sub.TGT}}$ that indicates an expected level of the output voltage $V_{\text{sub.OUT}}$. The voltage control circuit **34** then provides the target voltage $V_{\text{sub.TGT}}$ to each of the current regulators **30(1)-30(N)**.

[0037] FIG. **3** is a schematic diagram providing an exemplary illustration of the voltage control circuit **34**. Common elements between FIGS. **2** and **3** are shown therein with common element numbers and will not be re-described herein.

[0038] In an embodiment, the voltage control circuit **34** includes an error amplifier **36**, a voltage source **38**, and a voltage divider **40**. The voltage source **38** is configured to provide the reference

voltage V.sub.REF to the error amplifier **36**. The voltage divider **40**, which includes a pair of resistors R.sub.A and R.sub.B, is configured to provide a portion of the feedback voltage V.sub.OUT-FB to the error amplifier **36**. The error amplifier **36**, which is associated with a voltage error gain G.sub.EA, is configured to generate the target voltage V.sub.TGT as a function of the output current I.sub.OUT. More specifically, the error amplifier **36** can generate the target voltage V.sub.TGT in accordance with equation (Eq. 1) below.

[00001] $V_{TGT} = V_0 - R_{LL} * I_{OUT}$ (Eq. 1)

[0039] In the equation (Eq. 1), V.sub.0 represents a value of the output voltage V.sub.OUT when the output current I.sub.OUT is equal to zero, R.sub.LL represents a load line of the multi-phase voltage regulator **24**, and I.sub.OUT represents an expected level of the output current I.sub.OUT. The equation (Eq. 1) can be better visualized via a load line curve, as illustrated in FIG. 4.

[0040] With reference to FIG. 4, the load line R.sub.LL can be determined by the current sense gain G.sub.IS and the voltage error gain G.sub.EA, as shown in equation (Eq. 2) below.

[00002] $R_{LL} = G_{IS} / G_{EA}$ (Eq. 2)

[0041] With reference back to FIG. 2, because the target voltage V.sub.TGT is now a function of the output current I.sub.OUT, each of the current regulators **30(1)-30(N)** can thus extrapolate the output current I.sub.OUT from the target voltage V.sub.TGT. Accordingly, each of the current regulators **30(1)-30(N)** can self-regulate a respective one of the sub-currents I.sub.OUT-1-I.sub.OUT-N to be substantially identical to each of the other sub-currents I.sub.OUT-1-I.sub.OUT-N. In the context of the present disclosure, the sub-currents I.sub.OUT-1-I.sub.OUT-N are said to be substantially identical to one another when a respective difference between each of the sub-currents I.sub.OUT-1-I.sub.OUT-N is within plus-minus one percent ($\leq \pm 1\%$). As a result, it is possible to eliminate the current balancing circuit **22** and the set of third feedback loops Loop**3(1)**-Loop**3(N)** from the multi-phase voltage regulator **24**.

[0042] In an embodiment, each of the current regulators **30(1)-30(N)** includes a respective one of multiple controllers **42(1)-42(N)**, a respective one of multiple voltage converters **44(1)-44(N)**, a respective one of multiple first switching devices SA.sub.1-SA.sub.N, a respective one of multiple second switching devices SB.sub.1-SB.sub.N, and a respective one of multiple power inductors L.sub.1-L.sub.N. In an embodiment, each of the controllers **42(1)-42(N)** can be a pulse-width modulation (PWM) controller. Herein, each of the controllers **42(1)-42(N)** is configured to receive the target voltage V.sub.TGT from the voltage control circuit **34**. Each of the controllers **42(1)-42(N)** also receives a respective one of the sense currents I.sub.SNS-1-I.sub.SNS-N via a respective one of the set of first feedback loops Loop**1(1)**-Loop**1(N)**.

[0043] Each of the controllers **42(1)-42(N)** can thus extrapolate the expected level of the output current I.sub.OUT at the voltage output **26** from the target voltage V.sub.TGT and determine an expected level of the respective one of the sub-currents I.sub.OUT-1-I.sub.OUT-N. By comparing the expected level of the respective one of the sub-currents I.sub.OUT-1-I.sub.OUT-N against the present level of the respective one of the sub-currents I.sub.OUT-1-I.sub.OUT-N as indicated by the respective one of the sense currents I.sub.SNS-1-I.sub.SNS-N, each of the controllers **42(1)-42(N)** can then determine an adjustment value to the respective one of the sub-currents I.sub.OUT-1-I.sub.OUT-N. Subsequently, each of the controllers **42(1)-42(N)** can generate a respective one of the duty cycle signals **46(1)-46(N)**.

[0044] The voltage converters **44(1)-44(N)** can each be a buck converter, a boost converter, or a buck-boost converter. Each of the voltage converters **44(1)-44(N)** is configured to generate a respective one of multiple sub-voltages V.sub.OUT-1-V.sub.OUT-N in accordance with a respective one of the duty cycle signals **46(1)-46(N)**.

[0045] In a non-limiting example, each of the first switching devices SA.sub.1-SA.sub.N and the second switching devices SB.sub.1-SB.sub.N can be implemented by a semiconductor transistor, such as a metal-oxide semiconductor field-effect transistor (MOSFET). When each of the voltage

converters **44(1)-44(N)** is operating as the buck converter, only the first switching devices SA.sub.1-SA.sub.N are provided in the current regulators **30(1)-30(N)**. In contrast, when each of the voltage converters **44(1)-44(N)** is operating as the boost converter, only the second switching devices SB.sub.1-SB.sub.N are provided in the current regulators **30(1)-30(N)**. In case each of the voltage converters **44(1)-44(N)** is operating as the buck-boost converter, both the first switching devices SA.sub.1-SA.sub.N and the second switching devices SB.sub.1-SB.sub.N are provided in the current regulators **30(1)-30(N)**.

[0046] In an embodiment, each of the first switching devices SA.sub.1-SA.sub.N may be toggled based on a respective one of the duty cycle signals **46(1)-46(N)** to couple a respective one of the voltage converters **44(1)-44(N)** to a respective one of the power inductors L.sub.1-L.sub.N such that the respective one of the power inductors L.sub.1-L.sub.N can generate a respective one of the sub-currents I.sub.OUT-1-I.sub.OUT-N based on a respective one of the sub-voltages V.sub.OUT-1-V.sub.OUT-N. Likewise, each of the second switching devices SB.sub.1-SB.sub.N may also be toggled based on a respective one of the duty cycle signals **46(1)-46(N)** to couple a respective one of the power inductors L.sub.1-L.sub.N to the voltage output **26** such that the respective one of the power inductors L.sub.1-L.sub.N can provide a respective one of the sub-currents I.sub.OUT-1-I.sub.OUT-N to the voltage output **26**.

[0047] In addition to configuring each of the current regulators **30(1)-30(N)** to self-regulate the respective one of the sub-currents I.sub.OUT-1-I.sub.OUT-N to be substantially identical to each of the other sub-currents I.sub.OUT-1-I.sub.OUT-N, the multi-phase voltage regulator **24** may be required to report the output current I.sub.OUT to help control a corresponding electronic power system in accordance with performance and/or regulatory requirements. In this regard, the multi-phase voltage regulator **24** may further include a current determination circuit **48**. According to an embodiment of the present disclosure, the current determination circuit **48** can be configured to calculate the output current I.sub.OUT and report the output current I.sub.OUT to circuitries located outside the multi-phase voltage regulator **24**.

[0048] According to the equation (Eq. 1), the output current I.sub.OUT can be calculated based on the feedback V.sub.OUT-FB that indicates the present level of the output voltage V.sub.OUT in conjunction with a set of configured parameters (V.sub.0 and R.sub.LL). As described in FIG. 4, V.sub.0 represents the value of the output voltage V.sub.OUT when the output current I.sub.OUT is equal to zero and R.sub.LL represents the load line of the multi-phase voltage regulator **24**. In this regard, if the current determination circuit **48** is made aware of the set of configured parameters (V.sub.0 and R.sub.LL) and the feedback V.sub.OUT-FB of the output voltage V.sub.OUT, the current determination circuit **48** will then be able to calculate the output current I.sub.OUT in accordance with equation (Eq. 3) below.

[00003]
$$I_{OUT} = (V_{OUT-FB} - V_0) / R_{LL} \quad (\text{Eq. 3})$$

[0049] FIG. 5 is a schematic diagram of the current determination circuit **48** configured according to an embodiment of the present disclosure. Common elements between FIGS. 2 and 5 are shown therein with common element numbers and will not be re-described herein.

[0050] The current determination circuit **48** can include a configuration circuit **50**, a processing circuit **52**, and an analog-to-digital converter (ADC) **54**. The configuration circuit **50**, which can include such storage devices as register and flash memory, can be pre-configured to store the set of configured parameters (V.sub.0, R.sub.LL). The ADC **54** may be coupled to the second feedback loop Loop2 to receive the feedback V.sub.OUT-FB of the output voltage V.sub.OUT and convert the feedback V.sub.OUT-FB of the output voltage V.sub.OUT into a digital feedback voltage V.sub.OUT-D. The processing circuit **52**, which can be a general-purpose processor or an application-specific integrated circuit (ASIC), as an example, can be configured to calculate the output current I.sub.OUT in accordance with the equation (Eq. 3) above.

[0051] The multi-phase voltage regulator **24** of FIG. 2 can be provided in an electronic power

system to perform the functionalities described herein. In this regard, FIG. 6 is a schematic diagram of an exemplary electronic power system **100** wherein the multi-phase voltage regulator **24** of FIG. 2 can be provided.

[0052] In an embodiment, the electronic power system **100** includes a power source **102**, a conversion circuit **104**, a load circuit **106**, a feedback circuit **108**, and a control circuit **110**. The power source **102**, which can be an AC or a DC power source, is configured to generate an input voltage $V_{sub.IN}$ and/or an input current $I_{sub.IN}$.

[0053] The conversion circuit **104** is configured to convert the input voltage $V_{sub.IN}$ and/or the input current $I_{sub.IN}$ into an output voltage $V_{sub.OUT}$ and/or an output current $I_{sub.OUT}$. The conversion circuit **104** may be a step-down converter that converts a higher input voltage $V_{sub.IN}$ and/or a higher input current $I_{sub.IN}$ to a lower output voltage $V_{sub.OUT}$ and/or a lower output current $I_{sub.OUT}$. The conversion circuit **104** may be a step-up converter that converts a lower input voltage $V_{sub.IN}$ and/or a lower input current $I_{sub.IN}$ to a higher output voltage $V_{sub.OUT}$ and/or a higher output current $I_{sub.OUT}$. The conversion circuit **104** may also be a step-down and step-up converter that can toggle between step-down and step-up operations in accordance with a duty cycle to produce the output voltage $V_{sub.OUT}$ and/or the output current $I_{sub.OUT}$ at any level.

[0054] The load circuit **106** can be any type of electrical circuit, such as an electric vehicle (EV) motor, EV battery, power grid, data center server, and so on. The conversion circuit **104** is configured to provide the output voltage $V_{sub.OUT}$ and/or the output current $I_{sub.OUT}$ to the load circuit **106** via any suitable transmission medium.

[0055] The feedback circuit **108** is configured to provide various feedback to the control circuit **110**. As an example, the feedback circuit **108** can dynamically measure the output voltage $V_{sub.OUT}$ and/or the output current $I_{sub.OUT}$ being received by the load circuit **106** and report the measurement results to the control circuit **110**, either in real time or with hysteresis. The feedback circuit **108** may also monitor operating conditions (e.g., load impedance, operating frequency, thermal temperature, etc.) in the load circuit **106** and report such conditions to the control circuit **110**. The control circuit **110**, in turn, can dynamically control the conversion circuit **104** to adjust the output voltage $V_{sub.OUT}$ and/or the output current $I_{sub.OUT}$ based on the various feedback provided by the feedback circuit **108**.

[0056] In an embodiment, the multi-phase voltage regulator **24** may be configured to include the conversion circuit **104**, the feedback circuit **108**, and the control circuit **110**. In an embodiment, the multi-phase voltage regulator **24** may be integrated with the conversion circuit **104**. In an embodiment, the multi-phase voltage regulator **24** may be integrated with the control circuit **110**.

[0057] In an embodiment, the multi-phase voltage regulator **24** of FIG. 2 can be operated based on a process. In this regard, FIG. 7 is a flowchart of an exemplary process **200** whereby the multi-phase voltage regulator **24** of FIG. 2 can be configured to operate with fewer feedback loops.

[0058] Herein, the process **200** includes charging the output capacitor $C_{sub.OUT}$ by the output current $I_{sub.OUT}$, which includes the sub-currents $I_{sub.OUT-1}$ - $I_{sub.OUT-N}$ to provide the output voltage $V_{sub.OUT}$ at the voltage output **26** (step **202**). The process **200** also includes receiving the feedback voltage $V_{sub.OUT-FB}$ indicating the present level of the output voltage $V_{sub.OUT}$ at the voltage output **26** (step **204**). The process **200** also includes determining the target voltage $V_{sub.TGT}$ indicating the expected level of the output voltage $V_{sub.OUT}$ based on the feedback voltage $V_{sub.OUT-VB}$ and the reference voltage $V_{sub.REF}$ (step **206**). The process **200** also includes extrapolating the expected level of the output current $I_{sub.OUT}$ at the voltage output **26** from the target voltage $V_{sub.TGT}$ (step **208**). The process **200** also includes regulating each of the sub-currents $I_{sub.OUT-1}$ - $I_{sub.OUT-N}$ based on the expected level of the output current $I_{sub.OUT}$ such that the respective one of the sub-currents $I_{sub.OUT-1}$ - $I_{sub.OUT-N}$ is substantially identical to each of the other sub-currents $I_{sub.OUT-1}$ - $I_{sub.OUT-N}$ in the output current $I_{sub.OUT}$ (step **210**).

[0059] Those skilled in the art will recognize improvements and modifications to the embodiments of the present disclosure. All such improvements and modifications are considered within the scope of the concepts disclosed herein and the claims that follow.

Claims

1. A multi-phase voltage regulator comprising: an output capacitor coupled to a voltage output and charged by an output current comprising a plurality of sub-currents to provide an output voltage at the voltage output; a voltage control circuit configured to: receive a feedback voltage indicating a present level of the output voltage at the voltage output; and determine a target voltage indicating an expected level of the output voltage based on the feedback voltage and a reference voltage; and a plurality of sub-current generators each coupled to the voltage output and configured to: receive the target voltage from the voltage control circuit; extrapolate the expected level of the output current at the voltage output from the target voltage; and regulate the respective one of the plurality of sub-currents based on the expected level of the output current such that the respective one of the plurality of sub-currents is substantially identical to each other one of the plurality of sub-currents comprised in the output current.
2. The multi-phase voltage regulator of claim 1, wherein the voltage control circuit is further configured to determine the target voltage as a function of the output current.
3. The multi-phase voltage regulator of claim 2, wherein the target voltage is expressed as: $V_{\text{sub.TGT}} = V_{\text{sub.0}} - R_{\text{sub.LL}} \cdot I_{\text{sub.OUT}}$, wherein: $V_{\text{sub.TGT}}$ represents the target voltage; $V_{\text{sub.0}}$ represents the output voltage when the output current is zero; $R_{\text{sub.LL}}$ represents a load line of the multi-phase voltage regulator; and $I_{\text{sub.OUT}}$ represents the expected level of the output current.
4. The multi-phase voltage regulator of claim 1, further comprising: an output voltage feedback loop configured to provide the feedback voltage indicating the present level of the output voltage at the voltage output; and a plurality of current sensing circuits each configured to generate a respective one of a plurality of sense currents indicating the respective one of the plurality of sub-currents at the voltage output.
5. The multi-phase voltage regulator of claim 4, wherein each of the plurality of sub-current generators comprises a respective one of a plurality of current regulating circuits coupled to a respective one of the plurality of current sensing circuits and configured to: extrapolate the expected level of the output current at the voltage output from the target voltage; determine the expected level of the respective one of the plurality of sub-currents; compare the expected level of the respective one of the plurality of sub-currents against the present level of the respective one of the plurality of sub-currents as indicated by the respective one of the plurality of sense currents to determine an adjustment value to the respective one of the plurality of sub-currents; and adjust the respective one of the plurality of sub-currents to the expected level based on the determined adjustment value.
6. The multi-phase voltage regulator of claim 5, wherein each of the plurality of current regulating circuits comprises: a respective voltage converter configured to generate a respective sub-voltage based on the target voltage; and a respective power inductor configured to induce the respective one of the plurality of sub-currents based on the respective sub-voltage.
7. The multi-phase voltage regulator of claim 6, wherein the respective voltage converter is one of a buck converter, a boost converter, and a buck-boost converter.
8. The multi-phase voltage regulator of claim 6, wherein each of the plurality of current regulating circuits further comprises a respective switching device coupled between the respective voltage converter and the respective power inductor.
9. The multi-phase voltage regulator of claim 1, further comprising a current determination circuit configured to calculate the output current expressed as: $I_{\text{sub.OUT}} = (V_{\text{sub.0}} - V_{\text{sub.OUT}} -$

FB)/R.sub.LL, wherein: I.sub.OUT represents the output current calculated by the current determination circuit; V.sub.0 represents a level of the output voltage when the output current is equal to zero; V.sub.OUT-FB represents the feedback voltage indicating the present level of the output voltage at the voltage output; and R.sub.LL represents a load line of the multi-phase voltage regulator.

10. The multi-phase voltage regulator of claim 9, wherein the current determination circuit comprises: a configuration circuit configured to store the level of the output voltage when the output current is equal to zero and the load line of the multi-phase voltage regulator; and a processing circuit configured to calculate the output current based on the level of the output voltage when the output current is equal to zero, the load line of the multi-phase voltage regulator, and the feedback voltage indicating the present level of the output voltage at the voltage output.

11. A method for operating a multi-phase voltage regulator with fewer feedback loops comprising: charging an output capacitor by an output current comprising a plurality of sub-currents to provide an output voltage at a voltage output; receiving a feedback voltage indicating a present level of the output voltage at the voltage output; determining a target voltage indicating an expected level of the output voltage based on the feedback voltage and a reference voltage; extrapolating the expected level of the output current at the voltage output from the target voltage; and regulating each of the plurality of sub-currents based on the expected level of the output current such that the respective one of the plurality of sub-currents is substantially identical to each other one of the plurality of sub-currents comprised in the output current.

12. The method of claim 10, further comprising determining the target voltage as a function of the output current.

13. The method of claim 12, wherein determining the target voltage comprises determining the target voltage as being expressed as: $V_{sub.TGT} = V_{sub.0} - R_{sub.LL} * I_{sub.OUT}$, wherein: V.sub.TGT represents the target voltage; V.sub.0 represents the output voltage when the output current is zero; R.sub.LL represents the load line of the multi-phase voltage regulator; and I.sub.OUT represents the expected level of the output current.

14. The method of claim 11, further comprising: configuring an output voltage feedback loop to provide the feedback voltage indicating the present level of the output voltage at the voltage output; and configuring each of a plurality of current sensing circuits to generate a respective one of a plurality of sense currents indicating the respective one of the plurality of sub-currents at the voltage output.

15. The method of claim 14, further comprising: extrapolating the expected level of the output current at the voltage output from the target voltage; determining the expected level of the respective one of the plurality of sub-currents; comparing the expected level of the respective one of the plurality of sub-currents against the present level of the respective one of the plurality of sub-currents as indicated by the respective one of the plurality of sense currents to determine an adjustment value to the respective one of the plurality of sub-currents; and adjusting the respective one of the plurality of sub-currents to the expected level based on the determined adjustment value.

16. The method of claim 11, further comprising calculating the output current as expressed as: $I_{sub.OUT} = (V_{sub.0} - V_{sub.OUT-FB}) / R_{sub.LL}$, wherein: I.sub.OUT represents the output current as calculated; V.sub.0 represents a level of the output voltage when the output current is equal to zero; V.sub.OUT-FB represents the feedback voltage indicating the present level of the output voltage at the voltage output; and R.sub.LL represents a load line of the multi-phase voltage regulator.

17. An electronic power system comprising a conversion circuit coupled between a power source and a load circuit, the conversion circuit comprises a multi-phase voltage regulator comprising: an output capacitor coupled to a voltage output and charged by an output current comprising a plurality of sub-currents to provide an output voltage at the voltage output; a voltage control circuit configured to: receive a feedback voltage indicating a present level of the output voltage at the

voltage output; and determine a target voltage indicating an expected level of the output voltage based on the feedback voltage and a reference voltage; and a plurality of sub-current generators each coupled to the voltage output and configured to: receive the target voltage from the voltage control circuit; extrapolate the expected level of the output current at the voltage output from the target voltage; and regulate the respective one of the plurality of sub-currents based on the expected level of the output current such that the respective one of the plurality of sub-currents is substantially identical to each other one of the plurality of sub-currents comprised in the output current.

18. The electronic power system of claim 17, wherein the voltage control circuit is further configured to determine the target voltage as expressed as:

$V_{\text{sub.TGT}} = V_{\text{sub.0}} - R_{\text{sub.LL}} * I_{\text{sub.OUT}}$, wherein: $V_{\text{sub.TGT}}$ represents the target voltage; $V_{\text{sub.0}}$ represents the output voltage when the output current is zero; $R_{\text{sub.LL}}$ represents a load line of the multi-phase voltage regulator; and $I_{\text{sub.OUT}}$ represents the output current.

19. The electronic power system of claim 17, wherein the multi-phase voltage regulator further comprises: an output voltage feedback loop configured to provide the feedback voltage indicating the present level of the output voltage at the voltage output; and a plurality of current sensing circuits each configured to generate a respective one of a plurality of sense currents indicating the respective one of the plurality of sub-currents at the voltage output, wherein each of the plurality of sub-current generators comprises a respective one of a plurality of current regulating circuits coupled to a respective one of the plurality of current sensing circuits and configured to: extrapolate the expected level of the output current at the voltage output from the target voltage; determine the expected level of the respective one of the plurality of sub-currents; compare the expected level of the respective one of the plurality of sub-currents against the present level of the respective one of the plurality of sub-currents as indicated by the respective one of the plurality of sense currents to determine an adjustment value to the respective one of the plurality of sub-currents; and adjust the respective one of the plurality of sub-currents to the expected level based on the determined adjustment value.

20. The electronic power system of claim 17, wherein the multi-phase voltage regulator further comprises a current determination circuit configured to calculate the output current as expressed as: $I_{\text{sub.OUT}} = (V_{\text{sub.0}} - V_{\text{sub.OUT-FB}}) / R_{\text{sub.LL}}$, wherein: $I_{\text{sub.OUT}}$ represents the output current calculated by the current determination circuit; $V_{\text{sub.0}}$ represents a level of the output voltage when the output current is equal to zero; $V_{\text{sub.OUT-FB}}$ represents the feedback voltage indicating the present level of the output voltage at the voltage output; and $R_{\text{sub.LL}}$ represents a load line of the multi-phase voltage regulator.
